

Plenty of plastic

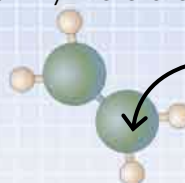
Bakelite proved so handy that chemists rushed to make new and even more useful plastics. In the 1930s, American chemist **WALLACE CAROTHERS** produced the *first completely human-made fiber*, nylon. In the same decade, newly invented Perspex was taking the place of glass and polythene began to be used in packaging. Soon, plastics were finding more and more uses.

Nylon stockings became a popular replacement for silk during the 1940s.

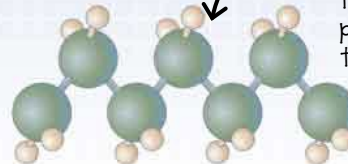


HOW IT WORKS

Plastics are polymers—materials made of molecules that consist of thousands or millions of atoms. Polymers are made by joining together small molecules (monomers) in a repeating structure that forms very long chains. Many different molecules can be used, and they can be combined in a variety of different ways—this is why there are so many types of plastic.



Ethene molecule (monomer)



Several ethene molecules join together to form polythene (a common type of plastic).

Going green

Today, plastic is everywhere—from the packaging of your food to the **RUBBER DUCK** in your bathtub. Most plastics are produced from oil, *a limited resource*, and can take centuries to decompose once discarded. **Bioplastics** made from organic material are now being developed as a **GREEN ALTERNATIVE** to oil-based plastics.

How it changed the world

Plastic is lightweight, cheap, tough, difficult to break, and doesn't rot. No wonder it's absolutely everywhere: in packaging, toys, furniture, computers, and clothing. But because it doesn't rot, garbage bins are full of plastic that will take hundreds of years to break down.

English inventor **ALEXANDER PARKES** made a semisynthetic plastic called **PARKESINE** in 1856, used to make a variety of domestic objects.

